UE26CY151A Engineering Chemistry · Unit 2 · Electrochemistry and Corrosion

Nernst Equation and its Derivation

Complete study pack for class 002: maximum work and free energy, the step-by-step derivation of the Nernst equation for a single electrode, the 0.0591/n form at 298 K, the effect of concentration and temperature, and the Nernst equation for a cell

Compiled from the PESU class 002 slide deck (labelled “Class 15 content”), the Nernst-equation part of the class 001 AV-summary transcript (the class 001 video covers classes 1 and 2), the consolidated Unit 2 slide PDF and past ESA papers. Generated 17 October 2026.

· How to use this pack

Topics 1–4 follow the lecture: free energy and cell potential, the derivation, using the single-electrode equation, and the cell form. Learn flowchart F2.1 for the 5–6-mark derivation; it reproduces the slides line by line. Then practise W4.1 (the ESA Jan 2025 cell) and W4.2. The cell numericals of class 006 all start from formula E4.1. Every topic has the same parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S002·4	Deck 002, slide 4 (rendered .pptx)
T001 00:21:42	Class 001 AV-summary transcript (covers classes 1 and 2), segment start
PDF p.13	Consolidated Unit 2 slide PDF (older syllabus), page 13
ESA Jan-25 s1 2a	Past paper: Jan 2025, set 1, question 2.a

Contents

1. 1 · Free energy and electrode potential
2. 2 · Derivation of the Nernst equation for a single electrode
3. 3 · Using the single-electrode equation: concentration and temperature
4. 4 · Nernst equation for a cell
5. Appendix A · Symbol glossary
6. Appendix B · Sources, coverage and gaps

1 Free energy and electrode potential

The maximum useful work a reversible electrode or cell can deliver equals the decrease in Gibbs free energy, $-\Delta G = W_{max}$. For an electrochemical system this work is the charge passed (nF) times the energy per unit charge (the potential E), so $\Delta G = -nFE$ and, under standard conditions, $\Delta G^\circ = -nFE^\circ$. A positive E therefore means a negative ΔG: a spontaneous reaction. These two relations are the starting point of the derivation.

Key terms and definitions

Maximum work. The maximum work that can be obtained from a system equals the decrease in free energy: $-\Delta G = W_{max}$. For an electrochemical system, W_{max} = total charge available × energy available per unit charge. S002·3 · T001 00:19:38 · T001 00:20:11

Faraday constant, F. The charge carried by one mole of electrons, 96500 C mol^{-1} on the slides. S002·3 · T001 00:20:21

> *"The maximum work that can be obtained is $-\Delta G = W_{max}$. For an electrochemical system, maximum work done is total charge available × energy available per unit charge. Therefore $W_{max} = nFE$; $\Delta G = -nFE$. Under standard conditions, $\Delta G^\circ = -nFE^\circ$."* S002·3

Formulas

E1.1 · Maximum work and free energy ESA ★★

$$-\Delta G = W_{max}$$

gives	Maximum useful work from a reversible process.
applies	Constant T and P, reversible operation.
units	J (or J mol^{-1})
note	Decrease in free energy = maximum work.

S002·3 · T001 00:19:38

E1.2 · Maximum electrical work ESA ★★★

$$W_{max} = \underbrace{nF}_{\text{charge}} \times \underbrace{E}_{\text{energy/charge}} = nFE$$

gives	Electrical work from n mol of electrons at potential E.
applies	Any electrode or cell.
units	C mol^{-1} × V = J mol^{-1}
note	1 V = 1 J C^{-1}.

S002·3 · T001 00:20:21 · T001 00:21:06

E1.3 · Free energy and potential ESA ★★★

$$\Delta G = -nFE, \qquad \Delta G^\circ = -nFE^\circ$$

GIVES	Links thermodynamics to potential.
APPLIES	Second form: standard conditions (1 M, 1 atm, 298 K).
UNITS	J mol^{-1}
NOTE	$E > 0 \Rightarrow \Delta G < 0$: spontaneous.

S002·3 · T001 00:16:25 · T001 00:21:24

Graphs and diagrams

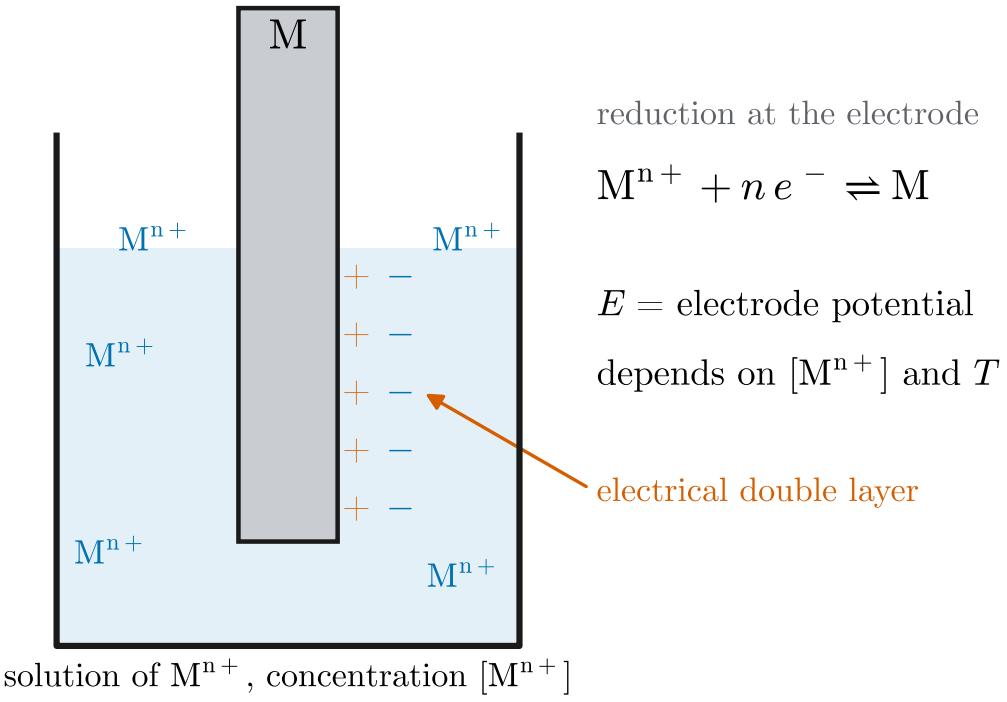
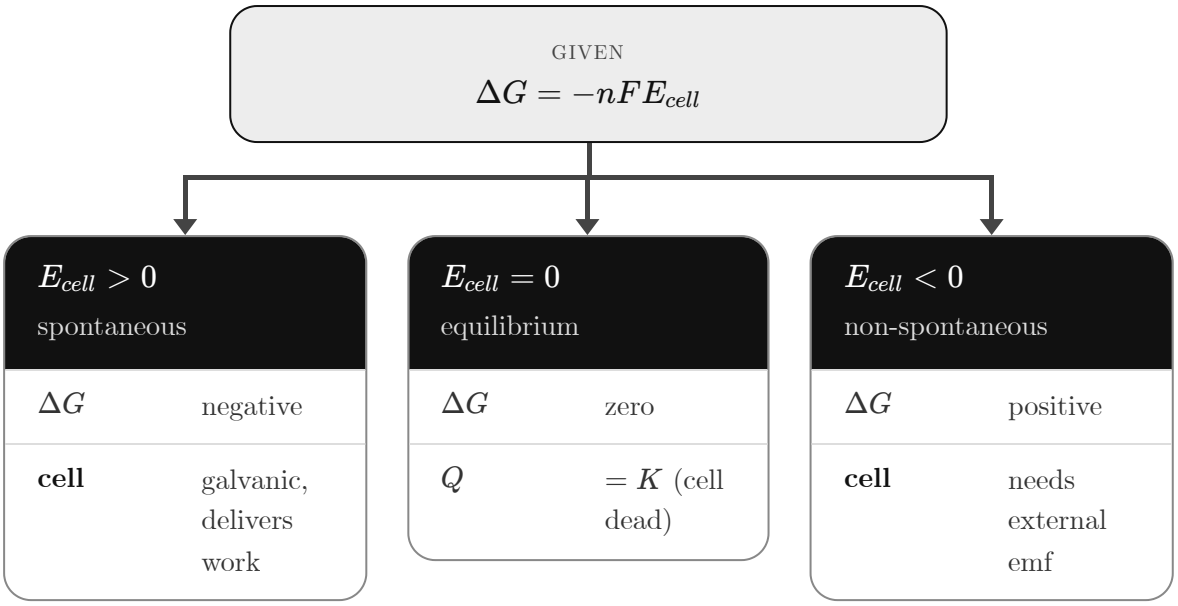


Figure G1.1. A single electrode: metal M in a solution of its ions. ESA ★★

ADDED FOR CLARITY S002·3 · T001 00:19:56

How to read it. The system of the derivation: a rod of metal M dipped in a solution of M^{n+}. The electrode reaction is written as a reduction. The charge separation at the interface (the electrical double layer from class 001) gives the electrode potential E, which the Nernst equation links to $[\mathrm{M}^{n+}]$ and T. Schematic, not to scale.

Flowcharts



Remark. The middle branch ($E = 0$ at equilibrium) is added for clarity; the lecture gives the two outer cases.

Figure F1.1. What the sign of E says about ΔG. ESA ★★

T001 00:16:39 · T001 00:16:51 · PDF p.10

Signs of ΔG, E and K

Reaction	ΔG	E_{cell}	K
Spontaneous	< 0	> 0	> 1
At equilibrium	0	0	$Q = K$
Non-spontaneous	> 0	< 0	< 1

ADDED FOR CLARITY T001 00:16:39 · S002·3

Important points and why

1. Total charge available = moles of electrons exchanged (n) × charge per mole of electrons (F) = nF; energy per unit charge = electrode potential E (volt = joule per coulomb). ESA ★★★ S002·3 · T001 00:20:21 · T001 00:20:50
2. $\Delta G = -nFE$: for a spontaneous reaction $\Delta G < 0$, so $E_{cell} > 0$; if $E_{cell} < 0$ the reaction is non-spontaneous. ESA ★★★ T001 00:16:39 · T001 00:16:51 · PDF p.10
3. E_{cell} is the driving force for the cell reaction. ESA ★★ T001 00:16:15 · PDF p.10

Why is $W_{max} = nFE$?
Work = charge × potential difference. n moles of electrons carry nF coulombs, and each coulomb gains E joules (1 V = 1 J C^{-1}).

S002·3 · T001 00:20:11

SHORT-NOTE RECAP

The maximum work obtainable equals the decrease in free energy, $-\Delta G = W_{max}$. For an electrochemical system, W_{max} = total charge (nF, $F = 96500$ C mol^{-1}) × energy per unit charge (E), so $W_{max} = nFE$, $\Delta G = -nFE$ and $\Delta G^\circ = -nFE^\circ$. A positive emf means negative ΔG: a spontaneous reaction.

2 Derivation of the Nernst equation for a single electrode

The Nernst equation is the quantitative relationship between the electrode potential and the concentration of the species with which the electrode is reversible. Starting from $\Delta G = \Delta G^\circ + RT\ln Q$ for the reduction $\mathrm{M^{n+}} + ne^- \rightleftharpoons \mathrm{M}$, substitute $\Delta G = -nFE$, $\Delta G^\circ = -nFE^\circ$ and $Q = [\mathrm{M}]/[\mathrm{M^{n+}}]$, divide by $-nF$, put $[\mathrm{M}] = 1$, convert to $\log_{10}$ and evaluate at 298 K. The result is $E = E^\circ + \frac{0.0591}{n}\log[\mathrm{M^{n+}}]$. This derivation is a 5–6-mark ESA question.

Key terms and definitions

Nernst equation. A quantitative relationship between electrode potential and concentration of species with which the electrode is reversible. S002·3 · T001 00:19:17

Reaction quotient, Q. The concentration of the products divided by the concentration of the reactants, each raised to its stoichiometric coefficient, at any instant (not only at equilibrium). S002·4 · T001 00:22:05 · T001 00:24:15

Standard electrode potential, E°. The electrode potential when all species are at unit concentration (1 M), gases at 1 atm, at 298 K. S002·4 · T001 00:19:00

> *"A thermodynamic equation which relates reaction quotient and decrease in free energy is given by $\Delta G = \Delta G^\circ + RT\ln Q$, where Q is the reaction quotient. The reaction quotient for the reaction is $Q = [\mathrm{M}]/[\mathrm{M^{n+}}]$."* S002·4

Formulas

E2.1 · Free energy and reaction quotient ESA ★★★

$$\Delta G = \Delta G^\circ + RT\ln Q$$

gives	How ΔG changes as Q changes.
applies	Any reaction; $\ln$ is the natural log.
units	J mol^{-1}
note	No nF here: $RT\ln Q$, not $\frac{RT}{nF}\ln Q$.

S002·4 · T001 00:21:42

E2.2 · Reaction quotient for the electrode

ESA ★★★

$$\mathrm{M^{n+}} + ne^- \rightleftharpoons \mathrm{M}, \qquad Q = \frac{[\mathrm{M}]}{[\mathrm{M^{n+}}]}$$

GIVES Q for the reduction.

APPLIES Electrode written as a reduction.

UNITS dimensionless

NOTE Product over reactant.

S002·4 · T001 00:22:05

E2.3 · After substituting and dividing by −nF

ESA ★★★

$$-nFE = -nFE^\circ + RT\ln\frac{[\mathrm{M}]}{[\mathrm{M^{n+}}]} \implies E = E^\circ - \frac{RT}{nF}\ln\frac{[\mathrm{M}]}{[\mathrm{M^{n+}}]}$$

GIVES The general Nernst equation, before simplification.

APPLIES Any T.

UNITS V

NOTE Dividing by $-nF$ flips the sign of the RT term.

S002·4 · T001 00:22:32 · T001 00:23:01

E2.4 · Single electrode, natural log

ESA ★★★

$$E = E^\circ + \frac{RT}{nF}\ln[\mathrm{M^{n+}}]$$

GIVES Electrode (reduction) potential at any T.

APPLIES $[\mathrm{M}] = 1$ for the pure metal.

UNITS V

NOTE $-\ln\frac{1}{x} = +\ln x$.

S002·5 · T001 00:23:17

E2.5 · Single electrode at 298 K

ESA ★★★

$$E = E^\circ + \frac{0.0591}{n}\log[\mathrm{M^{n+}}]$$

GIVES The form used in numericals.

APPLIES 298 K; reduction potential.

UNITS V

NOTE $0.0591 = 2.303 \times \frac{8.314\times 298}{96500}$.

S002·5 · T001 00:23:29 · T001 00:23:50

Flowcharts

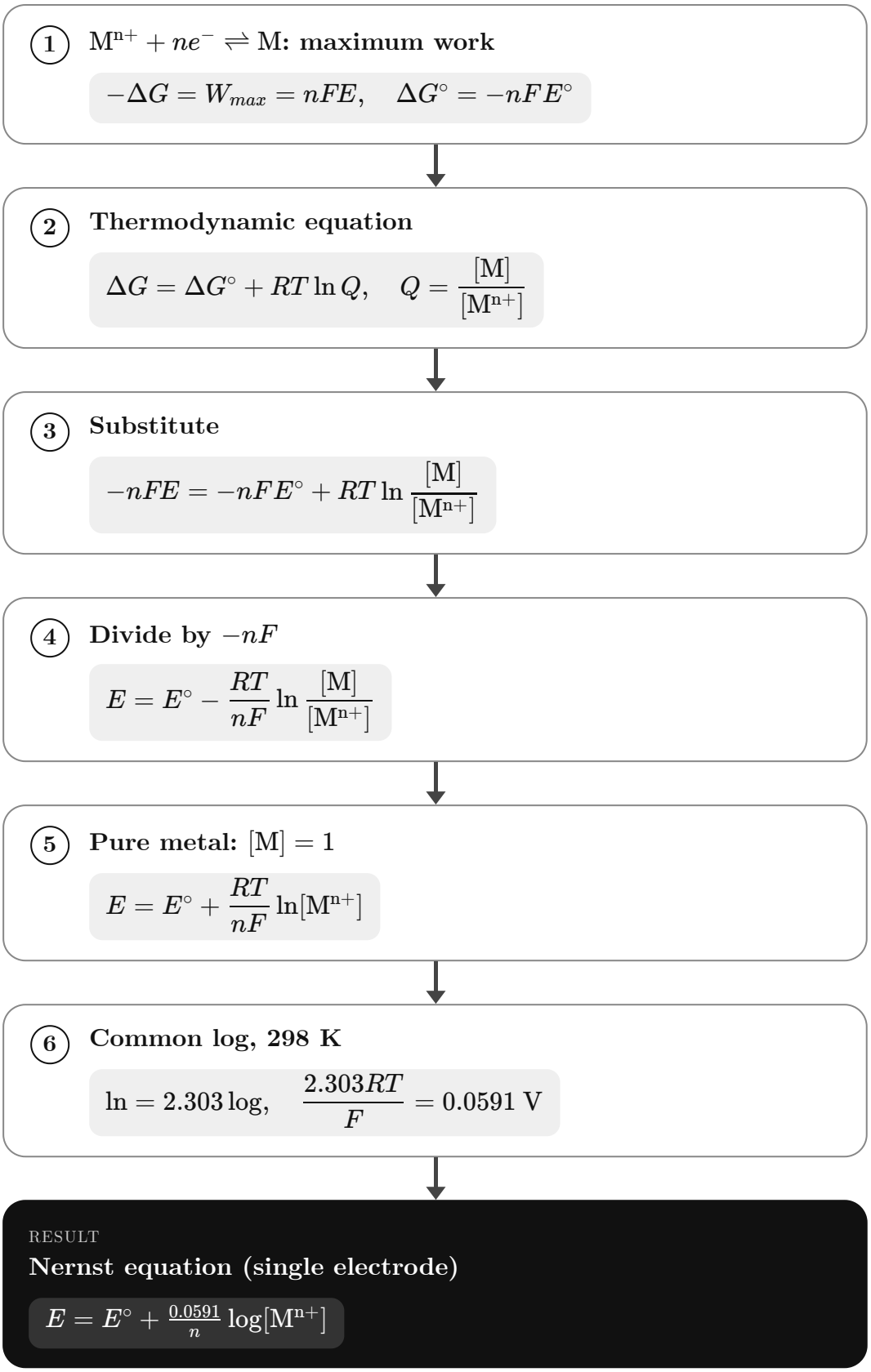


Figure F2.1. Derivation of the Nernst equation, step by step. ESA ★★★

S002·3 · S002·4 · S002·5 · T001 00:19:38 · T001 00:23:50 · ESA Dec-24 2a

Important points and why

1. The electrode reaction is written as a **reduction**, $\mathrm{M^{n+}} + ne^- \rightleftharpoons \mathrm{M}$, because electrode potentials are reduction potentials. ESA ★★★ S002·3 · T001 00:19:38
2. The system is a metal rod dipped in a solution of its own ions. ESA ★★ T001 00:19:56
3. Electrochemical series values are standard (E°: 1 M, 1 atm, 298 K), but real systems are seldom standard; the Nernst equation corrects E° for concentration, temperature and pressure. ESA ★★ T001 00:19:00 · T001 00:19:25
4. $[\mathrm{M}] = 1$ because the metal is a pure substance (pure solids have unit activity). ESA ★★★ S002·5 · T001 00:23:17
5. Natural log to common log: $\ln x = 2.303 \log x$; at 298 K, $2.303RT/F = 0.0591$ V. ESA ★★★ S002·5 · T001 00:23:29 · T001 00:23:50

Why does the sign change from $-\frac{RT}{nF}\ln\frac{[\mathrm{M}]}{[\mathrm{M^{n+}}]}$ to $+\frac{RT}{nF}\ln[\mathrm{M^{n+}}]$?
With $[\mathrm{M}] = 1$, $\ln\frac{1}{[\mathrm{M^{n+}}]} = -\ln[\mathrm{M^{n+}}]$, and the two minus signs make a plus.
S002·4 · S002·5

Why is the derivation done for the reduction reaction?
Standard electrode potentials are tabulated as reduction potentials, so writing $\mathrm{M^{n+}} + ne^- \to \mathrm{M}$ gives E on the same scale as E°.
T001 00:19:38

Exam trap. While extending to the cell the lecturer says "$\Delta G = \Delta G^\circ + \frac{RT}{nF}\ln Q$" (T001 00:24:39). That is wrong: the free-energy equation has $RT\ln Q$; the $1/nF$ appears only after dividing by $-nF$. ADDED FOR CLARITY T001 00:24:39 · S002·4

Exam trap. Sign: the single-electrode (reduction) form has $+\frac{0.0591}{n}\log[\mathrm{M^{n+}}]$; the cell form has $-\frac{0.0591}{n}\log Q$. Both are the same equation, since $Q = 1/[\mathrm{M^{n+}}]$ for the electrode. ADDED FOR CLARITY
S002·5

Worked examples

W2.1 · Model answer: derive the Nernst equation for a single electrode

ESA ★★★

given Electrode reaction $\mathrm{M^{n+}} + ne^- \rightleftharpoons \mathrm{M}$ (metal dipped in a solution of its ions). Derive E in terms of $[\mathrm{M^{n+}}]$.

step 1 Maximum work = decrease in free energy: $-\Delta G = W_{max}$.

step 2 W_{max} = total charge × energy per unit charge = $nF \times E$, so $\Delta G = -nFE$; under standard conditions $\Delta G^\circ = -nFE^\circ$.

step 3 Thermodynamics: $\Delta G = \Delta G^\circ + RT \ln Q$, with $Q = [\mathrm{M}]/[\mathrm{M^{n+}}]$.

step 4 Substitute: $-nFE = -nFE^\circ + RT \ln \frac{[\mathrm{M}]}{[\mathrm{M^{n+}}]}$.

step 5 Divide by $-nF$: $E = E^\circ - \frac{RT}{nF} \ln \frac{[\mathrm{M}]}{[\mathrm{M^{n+}}]}$.

step 6 $[\mathrm{M}] = 1$ (pure solid): $E = E^\circ + \frac{RT}{nF} \ln[\mathrm{M^{n+}}]$.

step 7 $\ln = 2.303 \log$; at 298 K with $R = 8.314$, $F = 96500$: $\frac{2.303RT}{F} = 0.0591$ V.

answer $$\boxed{E = E^\circ + \frac{0.0591}{n} \log[\mathrm{M^{n+}}] \text{ at 298 K}}$$

check At $[\mathrm{M^{n+}}] = 1$ M, $E = E^\circ$. Define every symbol (E°, n, R, T, F) for full marks; sympy: $2.303 \times 8.314 \times 298/96500 = 0.05913$.

ESA Dec-24 2a · S002·3 · S002·4 · S002·5 · T001 00:19:38

Past-paper questions

ESA Dec 2024 · 5 marks · UE23CY151A/B set 1, Q2.a · solved in W2.1

Derive the Nernst equation for a single electrode undergoing the following electrode reaction: $\mathrm{M^{n+}} + ne \rightarrow \mathrm{M}$.

short-note recap

For $\mathrm{M^{n+}} + ne^- \rightleftharpoons \mathrm{M}$: $\Delta G = \Delta G^\circ + RT \ln Q$ with $Q = [\mathrm{M}]/[\mathrm{M^{n+}}]$. Substituting gives $-nFE = -nFE^\circ + RT \ln \frac{[\mathrm{M}]}{[\mathrm{M^{n+}}]}$; dividing by $-nF$ and putting $[\mathrm{M}] = 1$ gives $E = E^\circ + \frac{RT}{nF} \ln[\mathrm{M^{n+}}]$. With $\ln = 2.303 \log$ at 298 K: $E = E^\circ + \frac{0.0591}{n} \log[\mathrm{M^{n+}}]$.

3 Using the single-electrode equation: concentration and temperature

At 298 K the electrode potential changes by $0.0591/n$ V for every tenfold change in $[\mathrm{M^{n+}}]$: diluting the ions makes a reduction potential more negative. A plot of E against $\log[\mathrm{M^{n+}}]$ is a straight line of slope $0.0591/n$ through E° at 1 M. The factor $2.303RT/F$ is proportional to T, so at another temperature it must be recomputed (0.0611 V at 308 K).

Formulas

E3.1 · The Nernst factor ESA ★★

$$\frac{2.303\,RT}{F} = \frac{2.303 \times 8.314 \times 298}{96500} = 0.0591\ \mathrm{V}$$

GIVES The 298 K constant.

APPLIES Slide constants; sympy gives 0.05913.

UNITS V

NOTE CODATA constants at 298.15 K give 0.05916 V; both round to 0.0591.

S002·5 · T001 00:23:40

E3.2 · Any temperature ESA ★

$$E = E^\circ + \frac{2.303\,RT}{nF}\log[\mathrm{M^{n+}}]$$

GIVES Electrode potential at temperature T.

APPLIES $T \neq 298$ K: recompute the factor (0.0611 V at 308 K).

UNITS V

NOTE Assumes E° is the 298 K value unless given.

ADDED FOR CLARITY S002·5 · T001 00:19:25

E3.3 · Change per decade ESA ★

$$\Delta E = \frac{0.0591}{n}\ \text{per tenfold change in } [\mathrm{M^{n+}}]$$

GIVES Slope of E vs $\log[\mathrm{M^{n+}}]$.

APPLIES 298 K.

UNITS V

NOTE $n = 1$: 59.1 mV; $n = 2$: 29.6 mV; $n = 3$: 19.7 mV.

ADDED FOR CLARITY S002·5

Graphs and diagrams

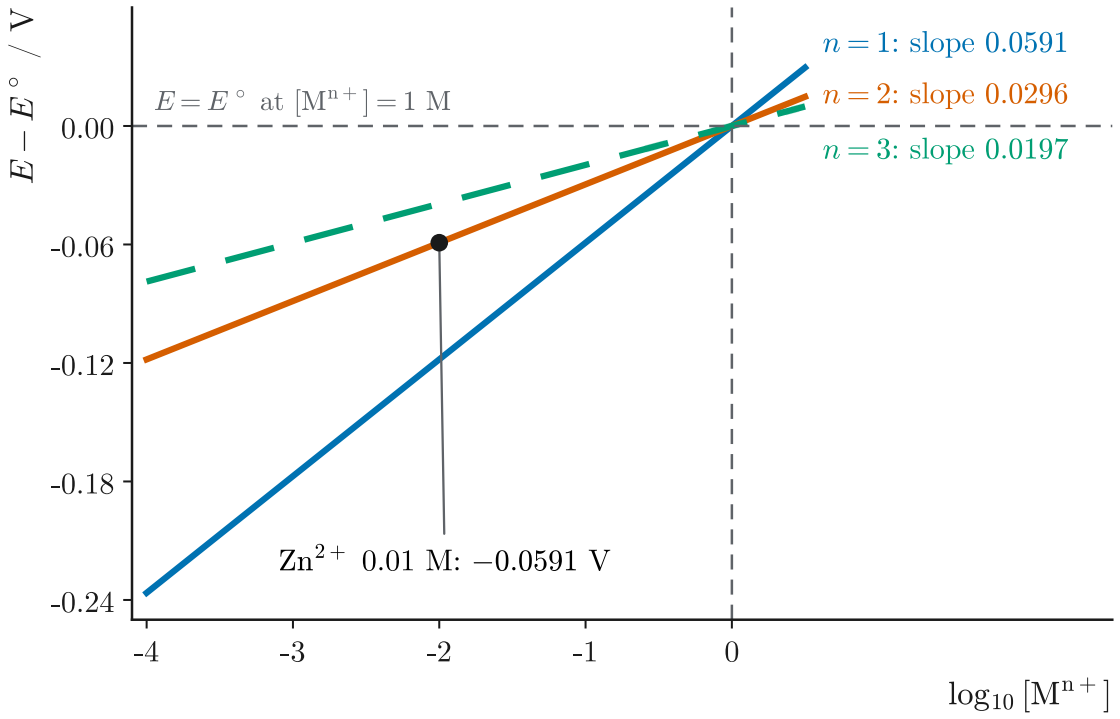


Figure G3.1. $E - E^\circ$ against $\log[M^{n+}]$ at 298 K. ESA ★★

ADDED FOR CLARITY S002·5

How to read it. Straight lines through the origin ($E = E^\circ$ at 1 M) with slope $0.0591/n$: 59.1 mV per decade for $n = 1$, 29.6 mV for $n = 2$, 19.7 mV for $n = 3$ (dashed). The marked point is Zn^{2+} at 0.01 M (two decades below 1 M, $n = 2$): -0.0591 V, as in W3.1.

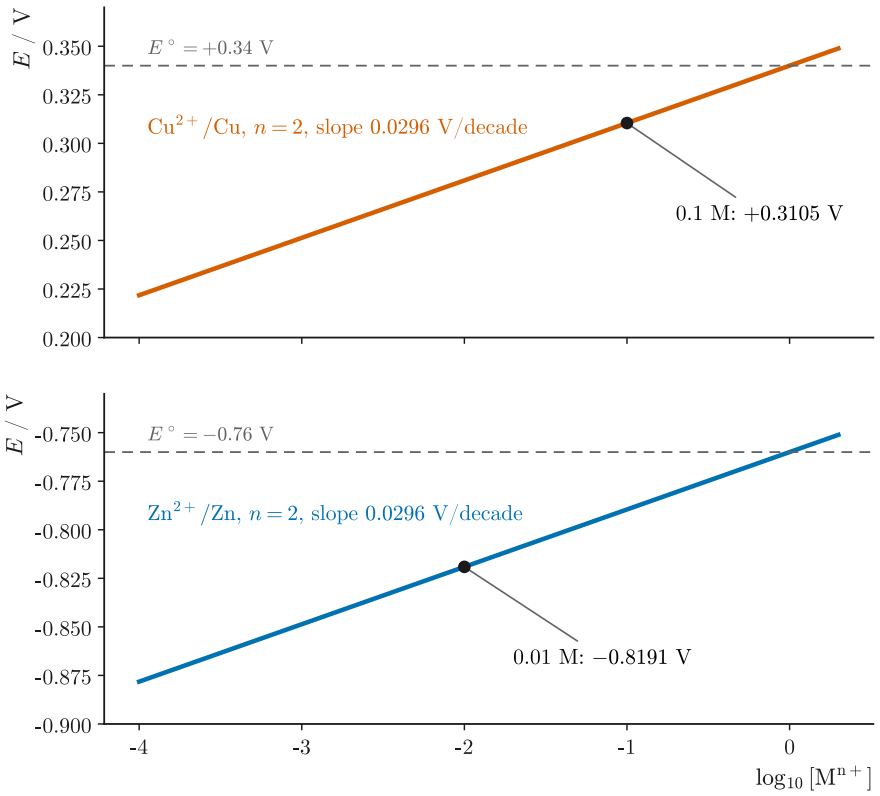


Figure G3.2. Electrode potentials of Cu and Zn against $\log[M^{n+}]$. ESA ★★

ADDED FOR CLARITY S002·5

How to read it. Each panel has its own vertical range (the lines are 1.10 V apart, which is E°_{cell} of the Daniell cell). Both have slope **0.0296** V per decade ($n = 2$). The grey dashed line is E°; the marked points are W3.2 (Cu, 0.1 M: +0.3105 V) and W3.1 (Zn, 0.01 M: −0.8191 V).

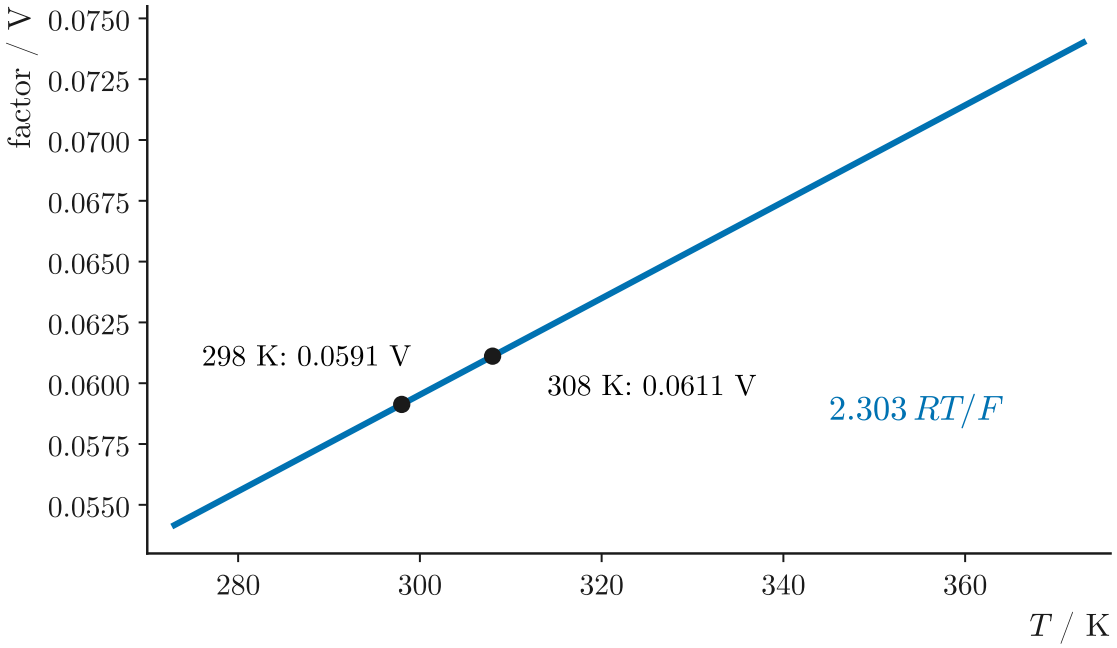
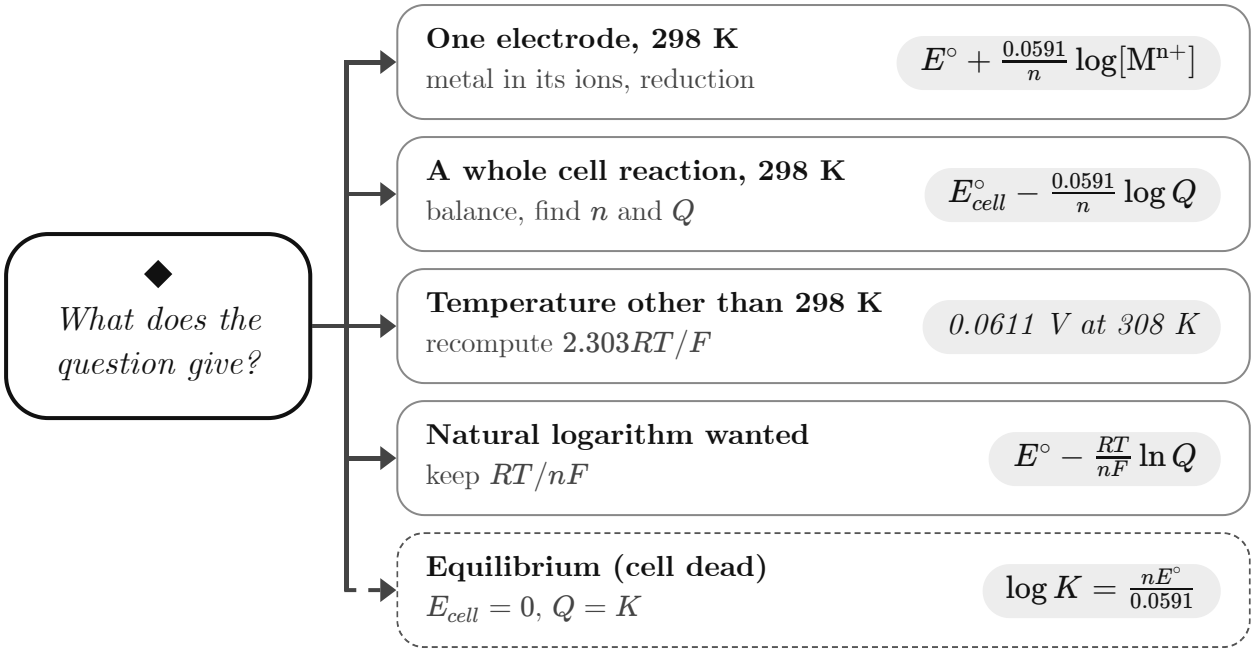


Figure G3.3. The Nernst factor $\mathbf{2.303}RT/F$ against temperature. ESA ★

ADDED FOR CLARITY S002·5 · T001 00:23:40

How to read it. The factor is proportional to T (slide constants): 0.0591 V at 298 K and 0.0611 V at 308 K. Use the value for the question's temperature; dividing by n gives the slope of G3.1.

Flowcharts



Remark. The dashed route (equilibrium constant) is added for clarity; it was not taught in class 002.

Figure F3.1. Which form of the Nernst equation? ESA ★★

S002·4 · S002·5

Important points and why

1. Raising $[\mathrm{M^{n+}}]$ raises E (the reduction becomes more favourable); diluting lowers it. Each tenfold change shifts E by $0.0591/n$ V. ESA ★★ ADDED FOR CLARITY S002·5
2. At $[\mathrm{M^{n+}}] = 1$ M, $\log 1 = 0$ and $E = E^\circ$: the Nernst equation reduces to the standard potential. ESA ★★ ADDED FOR CLARITY S002·5
3. The 0.0591 is valid only at 298 K. At another temperature use $2.303RT/F$ with that T (0.0611 V at 308 K). ESA ★★ S002·5 · T001 00:25:37

Exam trap. Do not use 0.0591 with ln: 0.0591 already contains the 2.303 and goes with $\log_{10}$; with ln use $RT/nF = 0.0257/n$. ADDED FOR CLARITY S002·5

Worked examples

W3.1 · Zinc electrode in 0.01 M Zn^{2+}

ESA ★★

given $Zn^{2+}(0.01\ \text{M})\,|\,Zn$, $E^\circ = -0.76$ V, 298 K.

step 1 $Zn^{2+} + 2e^- \rightleftharpoons Zn$, $n = 2$.

step 2 $E = -0.76 + \frac{0.0591}{2}\log(0.01) = -0.76 + 0.02955 \times (-2)$.

answer $\boxed{E = -0.8191\ \text{V}}$

check sympy: −0.81910 V; more negative than E°, as expected for dilute ions.

added for clarity S002·5

W3.2 · Copper electrode in 0.1 M Cu^{2+}

ESA ★★

given $Cu^{2+}(0.1\ \text{M})\,|\,Cu$, $E^\circ = +0.34$ V, 298 K.

step 1 $n = 2$.

step 2 $E = 0.34 + 0.02955\log(0.1) = 0.34 - 0.02955$.

answer $\boxed{E = +0.3105\ \text{V}}$

check sympy: 0.31045 V (one decade below 1 M: 29.6 mV lower).

added for clarity S002·5

W3.3 · Effect of temperature: zinc at 308 K

ESA ★

given $Zn^{2+}(0.01\ \text{M})\,|\,Zn$ at 308 K; take $E^\circ = -0.76$ V.

step 1 $\frac{2.303RT}{F} = \frac{2.303 \times 8.314 \times 308}{96500} = 0.06111$ V.

step 2 $E = -0.76 + \frac{0.06111}{2} \times (-2) = -0.76 - 0.06111$.

answer $\boxed{E = -0.8211\ \text{V}}$ (vs −0.8191 V at 298 K)

check sympy: −0.82111 V. Assumes E° unchanged with T, as such questions usually do.

added for clarity S002·5 · T001 00:19:25

short-note recap

E rises by $0.0591/n$ V per tenfold rise in $[M^{n+}]$ and equals E° at 1 M. Zn in 0.01 M Zn^{2+}: −0.8191 V; Cu in 0.1 M Cu^{2+}: +0.3105 V. The factor $2.303RT/F$ is proportional to T: 0.0591 V at 298 K, 0.0611 V at 308 K.

4 Nernst equation for a cell

The same steps applied to a whole cell reaction $aA + bB \rightleftharpoons cC + dD$ change only Q, which becomes products over reactants, each raised to its stoichiometric coefficient. At 298 K, $E_{cell} = E^\circ_{cell} - \frac{0.0591}{n}\log Q$, where n is the number of electrons in the balanced cell reaction. Pure solids are left out of Q. This form underlies every cell numerical in the UE23/UE24 papers.

Key terms and definitions

Standard emf, E°_{cell}. The emf of the cell when all concentrations are 1 M, all partial pressures 1 atm and the temperature 298 K. S002·5 · PDF p.10

> *"Nernst equation may also be used to calculate emf of electrochemical cells. For the cell reaction $aA + bB \rightleftharpoons cC + dD$, the Nernst equation is $E = E^\circ - \frac{0.0591}{n}\log\frac{[C]^c[D]^d}{[A]^a[B]^b}$; n = no. of electrons transferred, E°_{cell} = std. emf of the cell."* S002·5

Formulas

E4.1 · Nernst equation for a cell (298 K) ESA ★★★

$$E_{cell} = E^\circ_{cell} - \frac{0.0591}{n}\log\frac{[C]^c[D]^d}{[A]^a[B]^b}$$

gives	Cell emf at non-standard concentrations.
applies	$aA + bB \rightleftharpoons cC + dD$; n from the balanced equation.
units	V
note	General T: replace 0.0591 by $2.303RT/F$.

S002·5 · T001 00:24:15 · T001 00:25:06

E4.2 · Reaction quotient for a cell ESA ★★★

$$Q = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

gives	Products over reactants with coefficients as powers.
applies	Pure solids and liquids have activity 1 and are left out.
units	dimensionless
note	Al–Cu cell: $Q = [Al^{3+}]^2/[Cu^{2+}]^3$.

S002·5 · T001 00:24:15

E4.3 · Cell emf from electrode potentials

ESA ★★★

$$E_{cell} = E_{right} - E_{left} = E_{cathode} - E_{anode}$$

GIVES	Cell emf from two reduction potentials.
APPLIES	Both from E2.5 at their own concentrations.
UNITS	V
NOTE	Gives the same answer as E4.1.

T001 00:16:00 · PDF p.10

E4.4 · Equilibrium constant

ESA ★

$$E_{cell} = 0,\ Q = K \implies \log K = \frac{nE^\circ_{cell}}{0.0591}, \qquad \Delta G^\circ = -2.303\,RT \log K$$

GIVES	K from E°_{cell}.
APPLIES	At equilibrium (the cell is "dead").
UNITS	K dimensionless
NOTE	Not taught in class 002; a short note only.

ADDED FOR CLARITY NOT ON CURRENT SLIDES S002·5

Graphs and diagrams

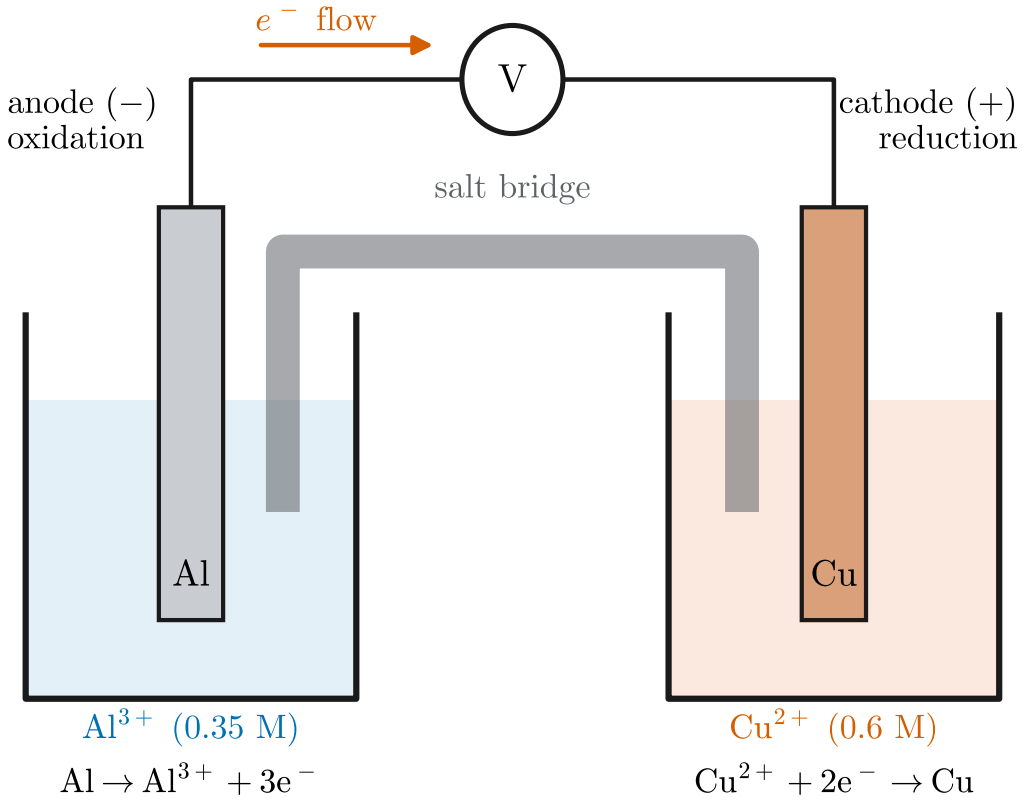


Figure G4.1. The Al–Cu cell of ESA Jan 2025. ESA ★★★

ADDED FOR CLARITY ESA Jan-25 s1 2a

How to read it. Al (lower E°, −1.66 V) is the anode and is oxidised; Cu (+0.34 V) is the cathode. Electrons flow through the wire from Al to Cu; the salt bridge completes the circuit. Cell: $\mathrm{Al}\,|\,\mathrm{Al^{3+}}(0.35\,\mathrm{M})\,\|\,\mathrm{Cu^{2+}}(0.6\,\mathrm{M})\,|\,\mathrm{Cu}$, $E_{cell} = 2.0024$ V. Schematic, not to scale.

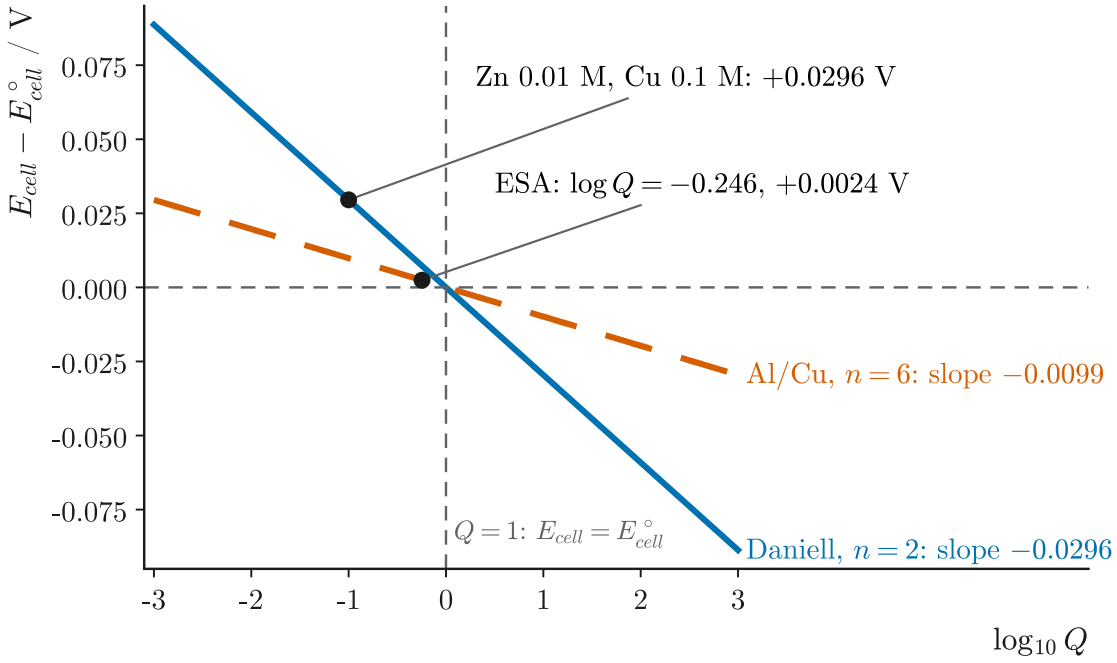
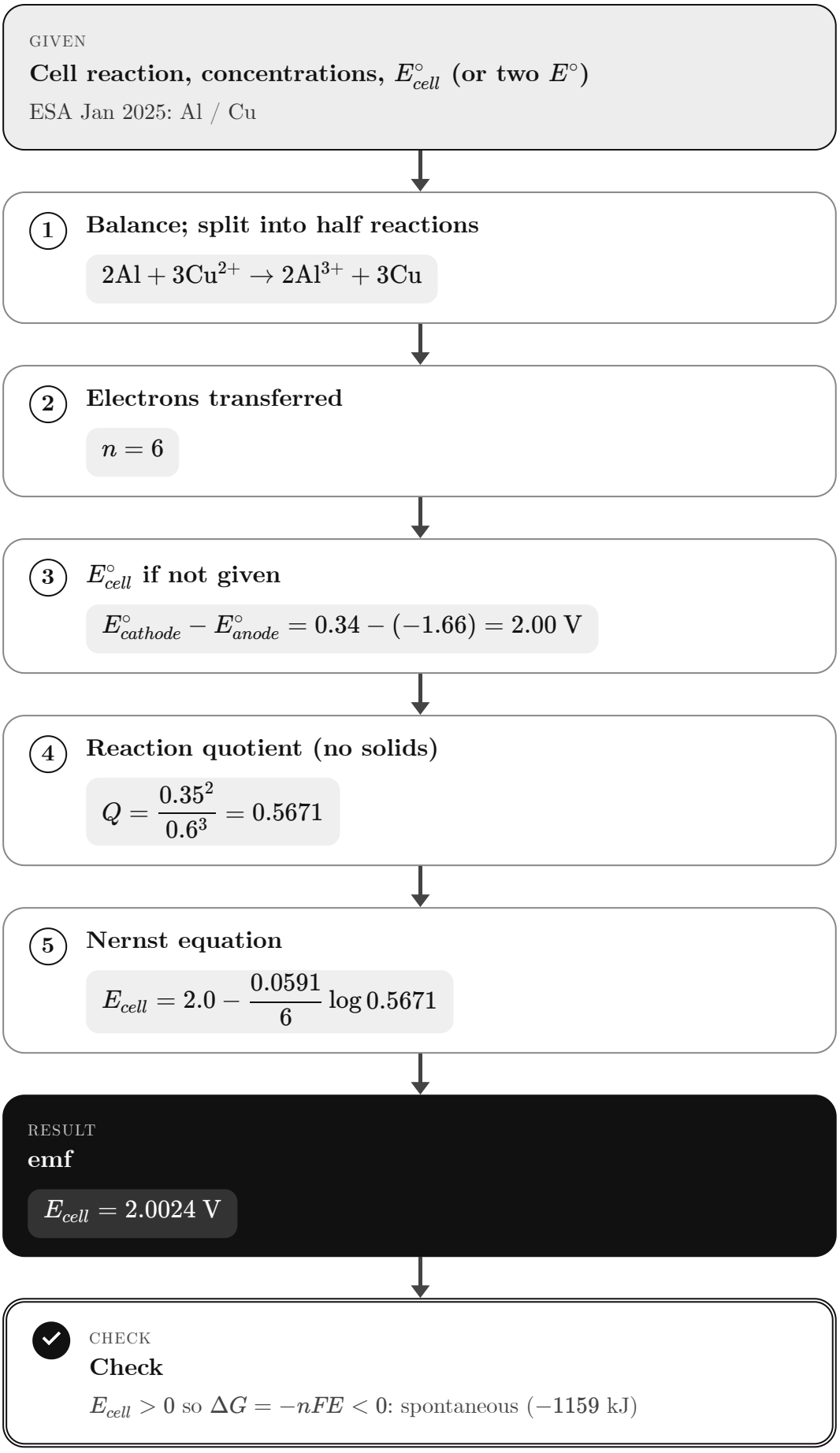


Figure G4.2. $E_{cell} - E^\circ_{cell}$ against $\log Q$. ESA ★★

ADDED FOR CLARITY S002·5 · ESA Jan-25 s1 2a

How to read it. Slope $-0.0591/n$: the larger n, the flatter the line, so the Al–Cu cell ($n = 6$, dashed) barely moves from 2.0 V. $Q < 1$ (left) raises the emf above E°_{cell}; $Q > 1$ lowers it. Points: W4.1 and W4.2.

Flowcharts


GIVEN
Cell reaction, concentrations, E°_{cell} (or two E°)
ESA Jan 2025: Al / Cu
1
Balance; split into half reactions
$2Al + 3Cu^{2+} \rightarrow 2Al^{3+} + 3Cu$
2
Electrons transferred
$n = 6$
3
E°_{cell} if not given
$E^\circ_{cathode} - E^\circ_{anode} = 0.34 - (-1.66) = 2.00\ V$
4
Reaction quotient (no solids)
$Q = \frac{0.35^2}{0.6^3} = 0.5671$
5
Nernst equation
$E_{cell} = 2.0 - \frac{0.0591}{6} \log 0.5671$
RESULT
emf
$E_{cell} = 2.0024\ V$
CHECK
Check
$E_{cell} > 0$ so $\Delta G = -nFE < 0$: spontaneous (-1159 kJ)

Remark. In a 6-mark question write the derivation (F2.1, cell form) before the calculation.

Exam trap. Balance first: the paper prints "+ Cu". Wrong n gives the wrong emf.

Figure F4.1. Solving a cell-emf numerical with the Nernst equation. ESA ★★★

S002·5 · ESA Jan-25 s1 2a

Nernst equation: single electrode vs cell

	Single electrode	Cell
Reaction	$\mathrm{M^{n+}} + ne^- \rightleftharpoons \mathrm{M}$	$a\mathrm{A} + b\mathrm{B} \rightleftharpoons c\mathrm{C} + d\mathrm{D}$
Q	$[\mathrm{M}]/[\mathrm{M^{n+}}] = 1/[\mathrm{M^{n+}}]$	$[\mathrm{C}]^c[\mathrm{D}]^d/[\mathrm{A}]^a[\mathrm{B}]^b$
E **means**	reduction (electrode) potential	emf E_{cell}
Standard value	E°	$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$
Form at 298 K	$E^\circ + \frac{0.0591}{n}\log[\mathrm{M^{n+}}]$	$E^\circ_{cell} - \frac{0.0591}{n}\log Q$
n	electrons in the half reaction	electrons in the balanced overall reaction
ESA	Dec 2024 Q2.a (5 marks)	Jan 2025 Q2.a (6 marks)

S002·4 · S002·5

Important points and why

1. For a cell only Q changes; $\Delta G = -nFE_{cell}$ and $\Delta G^\circ = -nFE^\circ_{cell}$ are substituted exactly as for one electrode. ESA ★★★ T001 00:24:15 · T001 00:24:52
2. n is the number of electrons in the **balanced** overall reaction: 2Al + $3\mathrm{Cu^{2+}}$ exchanges 6 electrons. ESA ★★★ ADDED FOR CLARITY ESA Jan-25 s1 2a
3. Solids (Al, Cu, Zn) do not appear in Q; only dissolved ions (and gas pressures) do. ESA ★★★ S002·5 · ESA Jan-25 s1 2a
4. The cell form gives the same emf as $E_{cathode} - E_{anode}$ with each electrode from E2.5 (check: Daniell 1.1296 V both ways). ESA ★★ ADDED FOR CLARITY T001 00:16:00

Why is the cell equation written with a minus sign and Q = products/reactants?
It comes straight from $\Delta G = \Delta G^\circ + RT\ln Q$: dividing $-nFE = -nFE^\circ + RT\ln Q$ by $-nF$ gives $E = E^\circ - \frac{RT}{nF}\ln Q$. More products (larger Q) lower the emf.
S002·5 · T001 00:24:52

Exam trap. ESA Jan 2025 prints "2Al + $3Cu^{2+}$ → $2Al^{3+}$ + Cu". Balance it (3Cu) and take $n = 6$; using $n = 2$ or $n = 3$ gives a wrong emf (2.0049 V for $n = 3$). ADDED FOR CLARITY ESA Jan-25 s1 2a

Exam trap. Raise each concentration to its coefficient: $[Al^{3+}]^2/[Cu^{2+}]^3$, not $0.35/0.6$.
ADDED FOR CLARITY S002·5 · ESA Jan-25 s1 2a

Worked examples

W4.1 · ESA: Al–Cu cell, derive and calculate the emf

ESA ★★★

GIVEN $2Al + 3Cu^{2+}(0.6\ M) \rightarrow 2Al^{3+}(0.35\ M) + 3Cu$ (paper prints "+ Cu"), $E^\circ_{cell} = 2.0$ V, 298 K.

STEP 1 Derivation: $\Delta G = \Delta G^\circ + RT \ln Q$; $-nFE_{cell} = -nFE^\circ_{cell} + RT \ln Q$; divide by $-nF$ and convert: $E_{cell} = E^\circ_{cell} - \frac{0.0591}{n} \log Q$.

STEP 2 Half reactions: $2Al \rightarrow 2Al^{3+} + 6e^-$, $3Cu^{2+} + 6e^- \rightarrow 3Cu$, so $n = 6$.

STEP 3 $Q = \frac{[Al^{3+}]^2}{[Cu^{2+}]^3} = \frac{0.35^2}{0.6^3} = \frac{0.1225}{0.216} = 0.5671$; $\log Q = -0.2463$.

STEP 4 $E_{cell} = 2.0 - \frac{0.0591}{6}(-0.2463) = 2.0 + 0.00243$.

ANSWER $\boxed{E_{cell} = 2.0024\ \text{V}}$

CHECK sympy: 2.00243 V (2.00243 with CODATA constants too). $E^\circ_{cell} = 0.34 - (-1.66) = 2.00$ V agrees with the given value. $\Delta G = -6 \times 96500 \times 2.0024 = -1159$ kJ.

ESA Jan-25 s1 2a · S002·5

W4.2 · Daniell cell at non-unit concentrations

ESA ★★

GIVEN $Zn\,|\,Zn^{2+}(0.01\ M)\,\|\,Cu^{2+}(0.1\ M)\,|\,Cu$, $E^\circ_{cell} = 1.10$ V, 298 K.

STEP 1 $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$, $n = 2$; $Q = \frac{0.01}{0.1} = 0.1$.

STEP 2 $E_{cell} = 1.10 - \frac{0.0591}{2} \log 0.1 = 1.10 + 0.02955$.

STEP 3 $\Delta G = -nFE_{cell} = -2 \times 96500 \times 1.1296$ J.

ANSWER $\boxed{E_{cell} = 1.1296\ \text{V};\ \Delta G = -218.0\ \text{kJ mol}^{-1}}$

CHECK Electrode route: $0.3105 - (-0.8191) = 1.1296$ V (W3.1, W3.2). $\Delta G^\circ = -212.3$ kJ. sympy agrees.

ADDED FOR CLARITY S002·5 · T001 00:16:00

W4.3 · Equilibrium constant of the Daniell cell ESA ★

GIVEN $E^\circ_{cell} = 1.10$ V, $n = 2$, 298 K.

STEP 1 At equilibrium $E_{cell} = 0$ and $Q = K$: $0 = E^\circ - \frac{0.0591}{n}\log K$.

STEP 2 $\log K = \frac{2\times 1.10}{0.0591} = 37.23$.

ANSWER $K \approx 1.7 \times 10^{37}$

CHECK sympy: log K = 37.225. The reaction goes essentially to completion.

ADDED FOR CLARITY · NOT ON CURRENT SLIDES · S002·5

Past-paper questions

ESA Jan 2025 · 6 marks · UE24CY151A set 1, Q2.a · solved in W4.1

Derive Nernst equation for the given cell and calculate the emf of the cell. $2Al + 3Cu^{2+}(0.6\,M) \rightarrow 2Al^{3+}(0.35\,M) + Cu$ (Given: $E^\circ_{cell} = 2.0$ V).

SHORT-NOTE RECAP

For $aA + bB \rightleftharpoons cC + dD$: $E_{cell} = E^\circ_{cell} - \frac{0.0591}{n}\log\frac{[C]^c[D]^d}{[A]^a[B]^b}$, n from the balanced reaction, solids omitted. ESA Jan 2025: 2Al + 3Cu^{2+}(0.6 M) → 2Al^{3+}(0.35 M) + 3Cu, $n = 6$, $Q = 0.5671$, $E_{cell} = 2.0024$ V.

A Symbol glossary

Symbol	Name	Meaning	Unit	Appears in
E	Electrode potential / cell emf	Reduction potential in E2.5; cell emf in E4.1 (same letter on the slides)	V	E1.2–E4.1
E°	Standard electrode potential	At 1 M, 1 atm, 298 K	V	E1.3, E2.3–E2.5
E°_{cell}	Standard emf	$E^\circ_{cathode} - E^\circ_{anode}$	V	E4.1, E4.4
ΔG, ΔG°	Free-energy change	Actual / standard	J mol^{-1}	E1.1, E1.3, E2.1
W_{max}	Maximum work	$= -\Delta G = nFE$	J	E1.1, E1.2
n	Electrons transferred	Moles of electrons in the balanced reaction	—	all
F	Faraday constant	96500 C mol^{-1}	C mol^{-1}	E1.2, E2.3
R	Gas constant	8.314 J K^{-1} mol^{-1}	J K^{-1} mol^{-1}	E2.1, E3.1
T	Temperature	In kelvin	K	E2.1, E3.2
Q	Reaction quotient	Products/reactants at any instant	—	E2.1, E2.2, E4.2
K	Equilibrium constant	Q at equilibrium	—	E4.4
[M^{n+}]	Ion concentration	Molar concentration of metal ions	mol L^{-1}	E2.2–E3.2
[M]	Metal	Pure solid: taken as 1	—	E2.2, E2.3
ln, log	Logarithms	Base e / base 10; $\ln x = 2.303 \log x$	—	E2.4, E2.5

B Sources, coverage and gaps

Source	Path	Use
Deck 002 (.pptx rendered at 80 dpi; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/002_Nernst_equation_and_its_derivation__49dac35211b6d743.*	6 slides; content on 3–5 (equations only in the images)
Transcript 001 (.transcript.md / .txt)	pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/001_Origin_of_electrode_potential_cell_potential__3e2916789bf0cc7b.*	Nernst part 00:16:15–00:25:55 (the video covers classes 1 and 2 per DOWNLOAD_MANIFEST)
Consolidated Unit 2 slides (8enterprise)	pesuc/First_Year_8enterprise/Files/Chemistry/Slides/Chem_Slides_Unit-2__1C1NjUHA4j.pdf	pp.12–14: same three slides, clean layout; confirms every equation
ESA papers (10 deduplicated)	pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/*/*.pdf	Dec 2024 Q2.a and Jan 2025 set 1 Q2.a (rendered at 90 dpi)
Class-work sheet	—	None exists for Chemistry in the bucket

Ten deduplicated Chemistry ESA papers (UE22CY151B Jul 2023 ×2, Dec 2023, May 2024; UE23CY151A/B Jan 2024 set 2, Jun 2024 ×2, Dec 2024, Dec 2024 UE22; UE24CY151A Jan 2025 sets 1 and 2) were checked. The Nernst derivation is asked directly twice: ESA Dec 2024 Q2.a (single electrode, 5 marks) and ESA Jan 2025 set 1 Q2.a (cell form plus emf, 6 marks). In addition, every UE23/UE24 paper has a Q2 cell numerical (calomel, redox, concentration cell) that uses the cell form of the equation; those are mapped to classes 004–006. The equilibrium constant, the temperature effect and the E vs log[M^{n+}] plot were never asked.

Known gaps, conflicts and corrections

1. There is no class 002 video or transcript. Per DOWNLOAD_MANIFEST the class 001 video (3e2916789bf0cc7b) is tagged for classes 1 and 2; its Nernst part (T001 00:18:50–00:25:55) is used and cited as T001.
2. No class-work (CW) sheet exists for Chemistry in the bucket, so all worked examples come from ESA papers or are constructed (flagged) and verified with sympy.
3. The deck is titled “Class 15 content” (older numbering); the content is the same as consolidated PDF pp.12–14.
4. Slide 002·3 renders with two overlapping text boxes (“W_{max} =” over “For an electrochemical system, maximum work done is”); a LibreOffice layout artefact, read from PDF p.12 instead. No content error.
5. The deck works no numerical example. Worked examples W3.1–W3.3, W4.2 and W4.3 are constructed with textbook E° values and flagged; W4.1 is ESA Jan 2025 set 1 Q2.a.
6. ESA Jan 2025 set 1 Q2.a prints “2Al + 3Cu^{2+}(0.6 M) → 2Al^{3+}(0.35 M) + Cu”: the copper is unbalanced. The balanced reaction has 3Cu and $n = 6$; the answer is $E_{cell} = 2.0024$ V (sympy). Solids do not enter Q, so the slip does not change the number.

7. The lecturer says "ΔG = ΔG° + RT over nF ln Q" at T001 00:24:39 while extending to the cell; the slide (and the single-electrode part, T001 00:21:42) has $\Delta G = \Delta G^\circ + RT \ln Q$. Corrected as an exam trap.
8. The lecturer says "RT log Q" for the natural log (T001 00:21:42) and the slide writes RTlnQ; the pack uses ln until the 2.303 conversion.
9. The slide value $F = 96500\ \mathrm{C\,mol^{-1}}$ is rounded (96485). With the slide constants $2.303RT/F = 0.05913$ V; with CODATA constants at 298.15 K it is 0.05916 V. Both round to the slide's 0.0591 V.
10. The equilibrium constant ($\log K = nE^\circ/0.0591$) and the temperature form are not on the slides or in the lecture; they appear as short notes flagged "added for clarity".
11. Transcript recognition errors were corrected silently: "learnst" and "nonst" → Nernst, "Faradin" → Faraday, "stechometric"/"stepometric" → stoichiometric, "R P over N F" → RT/nF, "P is the temperature" → T.

Coverage

Source	Citations	Slides cited (slide × citations)
S002	62	3×14, 4×12, 5×36
ESA	10	
PDF	5	
T001	55	

Deck 002 has six slides: 1 (title), 2 (class content), 3–5 (content) and 6 (thank you). LibreOffice overlaps two text boxes on slide 3 (the W_{max} line); the same slide in the consolidated PDF (p.12) is clean and identical in content, so PDF pp.12–14 were used to confirm every equation. There is no separate class 002 video: the class 001 video (T001) teaches the derivation from 00:18:50 to 00:25:55, and that part is cited as T001.